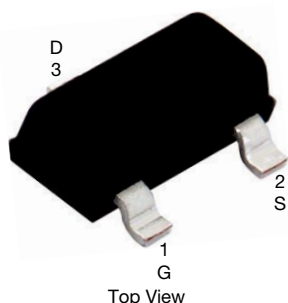


N-Channel 20 V (D-S) MOSFET

SOT-23 (TO-236)



Top View

Marking code: N2

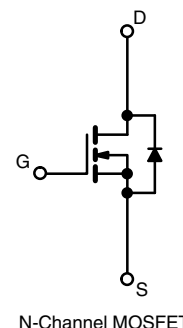
FEATURES

- TrenchFET® power MOSFET
- Material categorization:
for definitions of compliance please see
www.vishay.com/doc?99912


RoHS
COMPLIANT
HALOGEN
FREE
Available

APPLICATIONS

- Load switching for portable devices
- DC/DC converter



N-Channel MOSFET

PRODUCT SUMMARY

V_{DS} (V)	20
$R_{DS(on)}$ max. (Ω) at $V_{GS} = 4.5$ V	0.057
$R_{DS(on)}$ max. (Ω) at $V_{GS} = 2.5$ V	0.075
Q_g typ. (nC)	3.5
I_D (A)	2.9
Configuration	Single

ORDERING INFORMATION

Package	SOT-23
Lead (Pb)-free	Si2302CDS-T1-E3
Lead (Pb)-free and halogen-free	Si2302CDS-T1-GE3

ABSOLUTE MAXIMUM RATINGS ($T_A = 25$ °C, unless otherwise noted)

PARAMETER	SYMBOL	5 S	STEADY STATE	UNIT
Drain-source voltage	V_{DS}	20	20	V
Gate-source voltage	V_{GS}	± 8	± 8	
Continuous drain current ($T_J = 150$ °C) ^a	I_D	$T_A = 25$ °C	2.9	A
		$T_A = 70$ °C	2.3	
Pulsed drain current ^b	I_{DM}	10	10	
Continuous source current (diode conduction) ^a	I_S	0.72	0.6	
Power dissipation ^a	P_D	$T_A = 25$ °C	0.86	W
		$T_A = 70$ °C	0.55	
Operating junction and storage temperature range	T_J, T_{stg}	-55 to +150	-55 to +150	°C

THERMAL RESISTANCE RATINGS

PARAMETER	SYMBOL	TYPICAL	MAXIMUM	UNIT
Maximum junction-to-ambient ^a	R_{thJA}	$t \leq 5$ s	120	°C/W
		Steady state	140	
Maximum junction-to-foot	R_{thJF}	62	78	

Notes

- a. Surface mounted on 1" x 1" FR4 board
b. Pulse width limited by maximum junction temperature



SPECIFICATIONS (T _A = 25 °C, unless otherwise noted)						
PARAMETER	SYMBOL	TEST CONDITIONS	LIMITS			UNIT
			MIN.	TYP.	MAX.	
Static						
Drain-source breakdown voltage	V _{DS}	V _{GS} = 0 V, I _D = 250 μA	20	-	-	V
Gate-threshold voltage	V _{GS(th)}	V _{DS} = V _{GS} , I _D = 250 μA	0.40	-	0.85	
Gate-body leakage	I _{GSS}	V _{DS} = 0 V, V _{GS} = ± 8 V	-	-	± 100	nA
Zero gate voltage drain current	I _{DSS}	V _{DS} = 20 V, V _{GS} = 0 V	-	-	0.1	μA
		V _{DS} = 20 V, V _{GS} = 0 V, T _J = 50 °C	-	-	4	
		V _{DS} = 20 V, V _{GS} = 0 V, T _J = 70 °C	-	-	15	
On-state drain current ^a	I _{D(on)}	V _{DS} ≥ 10 V, V _{GS} = 4.5 V	6	-	-	A
Drain-source on-resistance ^a	R _{DS(on)}	V _{GS} = 4.5 V, I _D = 3.6 A	-	0.045	0.057	Ω
		V _{GS} = 2.5 V, I _D = 3.1 A	-	0.056	0.075	
Forward transconductance ^a	g _{fs}	V _{DS} = 5 V, I _D = 3.6 A	-	13	-	S
Diode forward voltage	V _{SD}	I _S = 0.95 A, V _{GS} = 0 V	-	0.7	1.2	V
Dynamic ^b						
Total gate charge	Q _g	V _{DS} = 10 V, V _{GS} = 4.5 V, I _D = 3.6 A	-	3.5	5.5	nC
Gate-source charge	Q _{gs}		-	0.6	-	
Gate-drain charge	Q _{gd}		-	0.45	-	
Gate resistance	R _g	f = 1 MHz	2	4	8	Ω
Switching						
Turn-on delay time	t _{d(on)}	V _{DD} = 10 V, R _L = 2.78 Ω I _D ≅ 3.6 A, V _{GEN} = 4.5 V, R _g = 1 Ω	-	8	15	ns
Rise time	t _r		-	7	15	
Turn-off delay time	t _{d(off)}		-	30	45	
Fall time	t _f		-	7	15	
Source-drain reverse recovery time	t _{rr}	I _F = 3.6 A, di/dt = 100 A/μs	-	8.5	15	nC
Body diode reverse recovery charge	Q _{rr}		-	2	4	

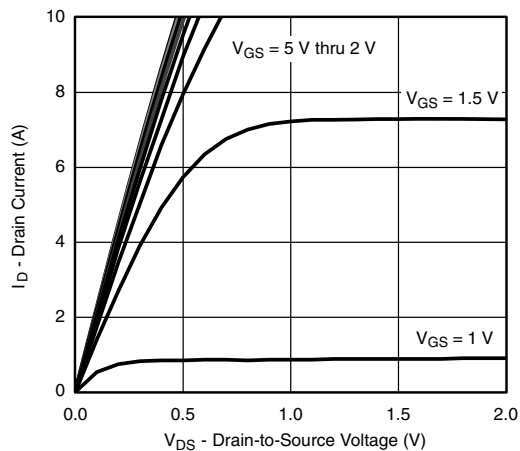
Notes

- a. Pulse test: Pulse width $\leq 300\text{ }\mu\text{s}$, duty cycle $\leq 2\%$
b. Guaranteed by design, not subject to production testing

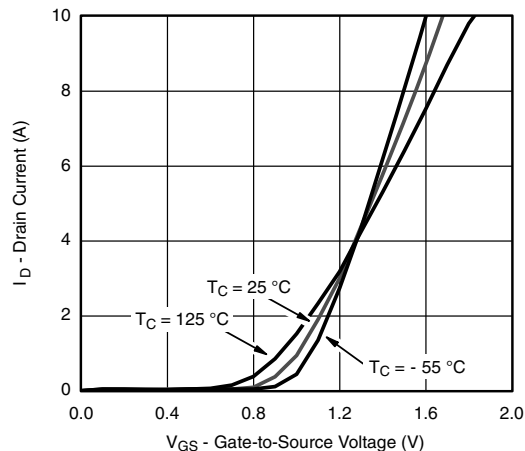
Stresses beyond those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated in the operational sections of the specifications is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.



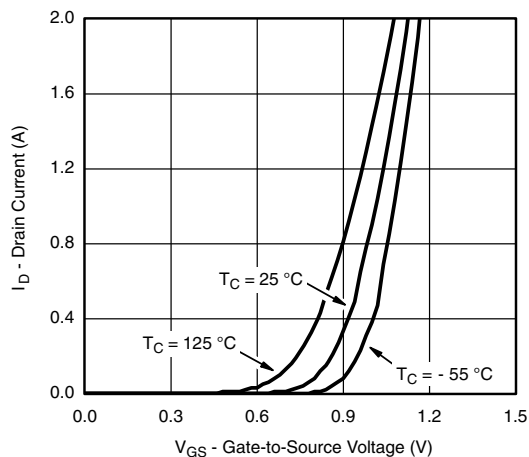
TYPICAL CHARACTERISTICS (25 °C, unless otherwise noted)



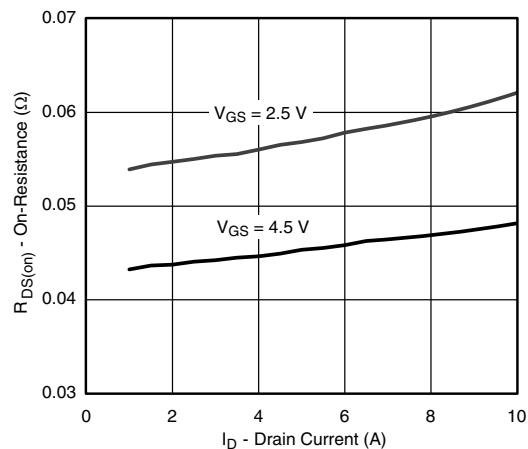
Output Characteristics



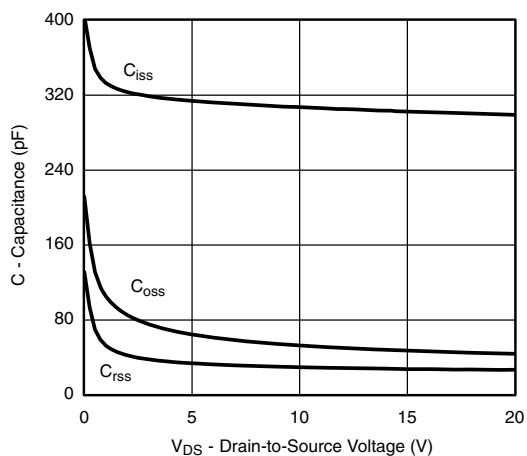
Transfer Characteristics



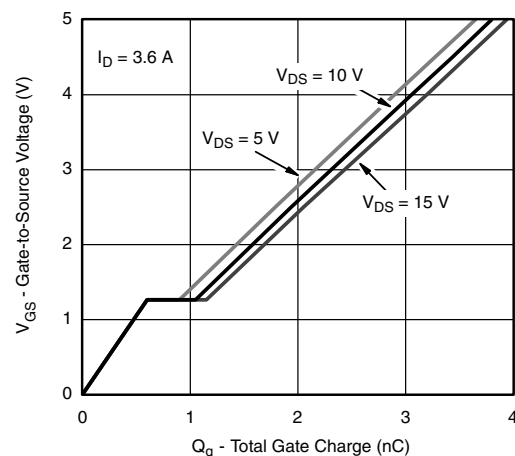
Transfer Characteristics



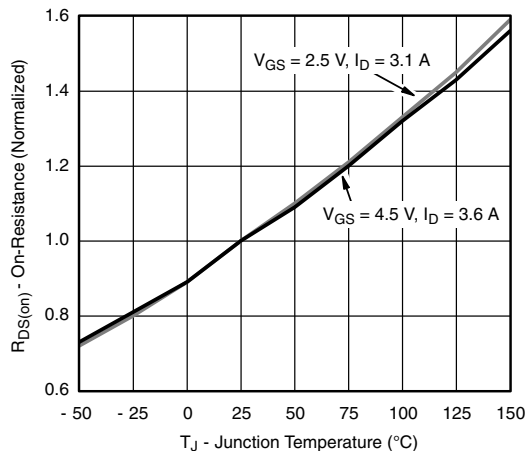
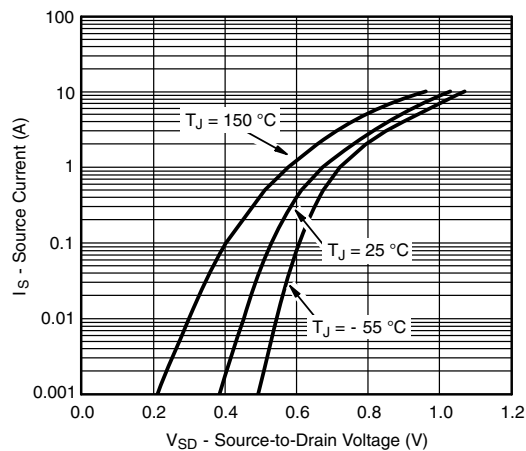
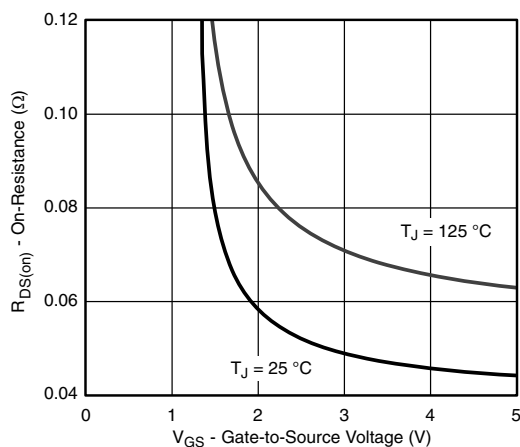
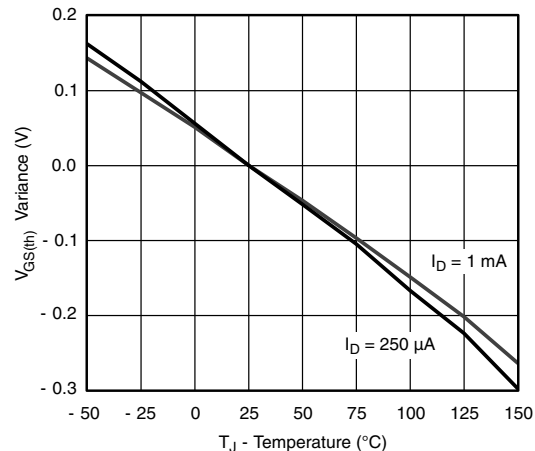
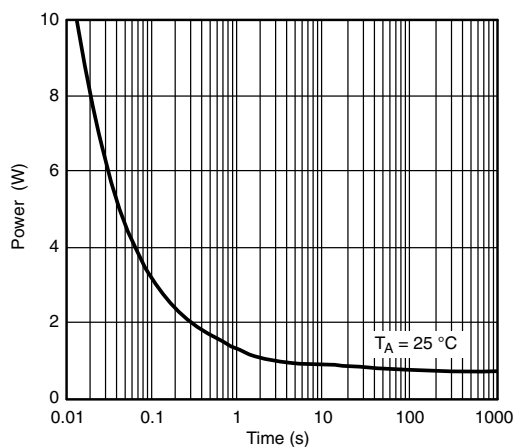
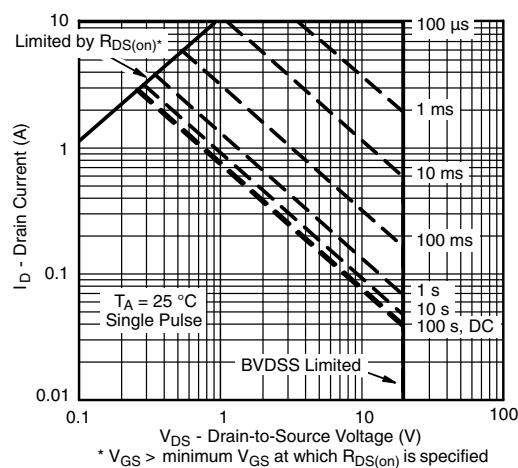
On-Resistance vs. Drain Current



Capacitance

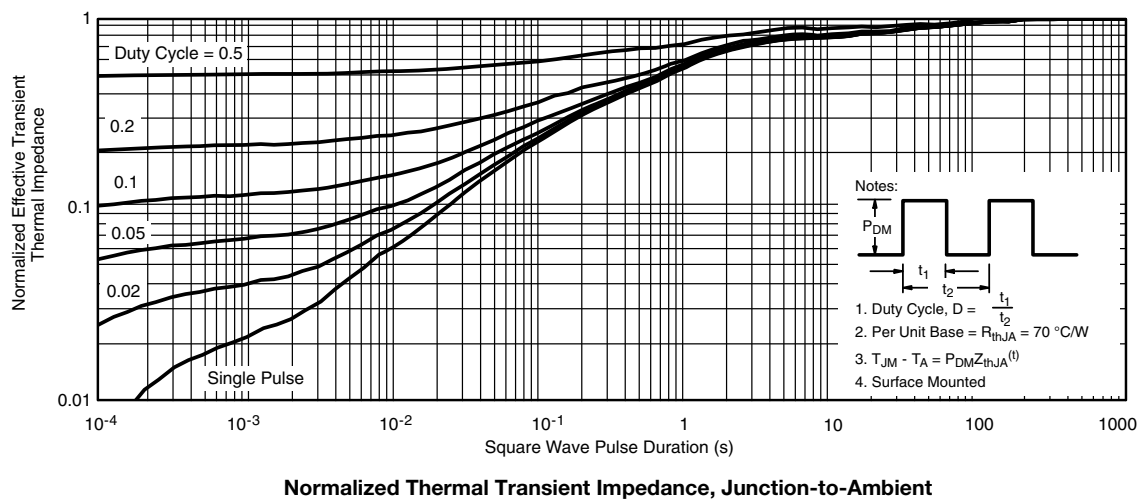


Gate Charge

TYPICAL CHARACTERISTICS (25 °C, unless otherwise noted)

On-Resistance vs. Junction Temperature

Source-Drain Diode Forward Voltage

On-Resistance vs. Gate-to-Source Voltage

Threshold Voltage

Single Pulse Power

Safe Operating Area, Junction-to-Ambient

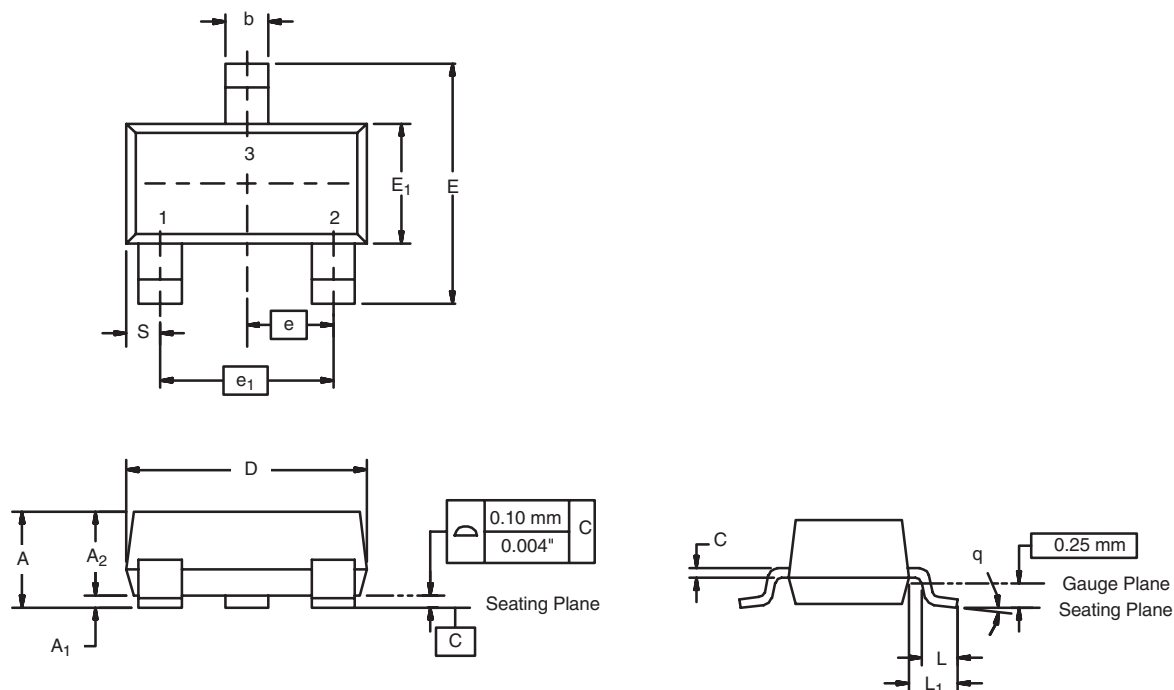


TYPICAL CHARACTERISTICS (25 °C, unless otherwise noted)



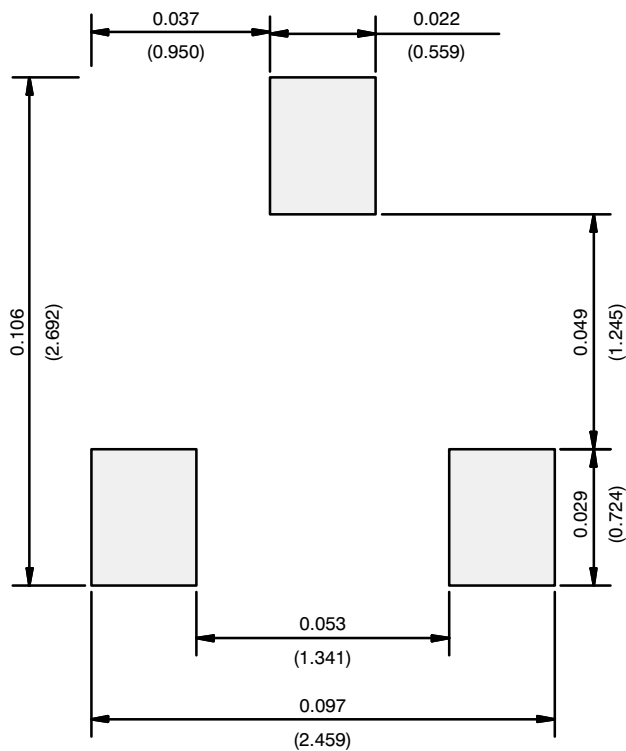
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SOT-23 (TO-236): 3-LEAD



Dim	MILLIMETERS		INCHES	
	Min	Max	Min	Max
A	0.89	1.12	0.035	0.044
A ₁	0.01	0.10	0.0004	0.004
A ₂	0.88	1.02	0.0346	0.040
b	0.35	0.50	0.014	0.020
c	0.085	0.18	0.003	0.007
D	2.80	3.04	0.110	0.120
E	2.10	2.64	0.083	0.104
E ₁	1.20	1.40	0.047	0.055
e	0.95 BSC		0.0374 Ref	
e ₁	1.90 BSC		0.0748 Ref	
L	0.40	0.60	0.016	0.024
L ₁	0.64 Ref		0.025 Ref	
S	0.50 Ref		0.020 Ref	
q	3°	8°	3°	8°
ECN: S-03946-Rev. K, 09-Jul-01				
DWG: 5479				

RECOMMENDED MINIMUM PADS FOR SOT-23



Recommended Minimum Pads
Dimensions in Inches/(mm)

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